

Comprehensive Sentiment Analysis Model Evaluation

Model Performance Ranking

Rank	Model	Pearson	Spearman	MAE	RMSE	Accuracy	Macro F1	Weighted F1	Overall Score
1	llama_3_0.650b_versatile	0.702	0.625	0.207	0.316	0.680	0.682	0.678	0.877
2	claude_3_5_sonnet_single	0.703	0.692	0.205	0.290	0.643	0.643	0.623	0.835
3	gpt_4o	0.731	0.716	0.196	0.282	0.635	0.630	0.606	0.834
4	gemma2_9b_it	0.608	0.609	0.242	0.357	0.690	0.692	0.684	0.832
5	gemini_2_flash_single	0.683	0.676	0.201	0.277	0.639	0.638	0.625	0.827
6	deepseek_chat	0.716	0.684	0.182	0.259	0.600	0.587	0.554	0.742
7	lsd_en	0.531	0.520	0.238	0.343	0.625	0.629	0.625	0.646
8	mixtral_8x7b_32768	0.738	0.650	0.268	0.363	0.590	0.579	0.563	0.589
9	lsd_fr	0.305	0.319	0.253	0.357	0.500	0.487	0.501	0.191
10	deepseek_r1_distill_llama_70b	0.173	0.172	0.405	0.540	0.500	0.501	0.500	0.121

Visual Performance Comparison

Rank	Model	Correlation	Error	Classification	Overall
1	llama_3_3_70b_versatile	0.652	0.207	0.682	0.877
2	claude_3_5_sonnet_single	0.703	0.205	0.643	0.835
3	gpt_4o	0.731	0.196	0.630	0.834
4	gemma2_9b_it	0.608	0.242	0.692	0.832
5	gemini_2_flash_single	0.683	0.201	0.638	0.827
6	deepseek_chat	0.716	0.182	0.587	0.742
7	lsd_en	0.531	0.238	0.629	0.646
8	mixtral_8x7b_32768	0.738	0.268	0.579	0.589
9	lsd_fr	0.305	0.253	0.487	0.191
10	deepseek_r1_distill_llama_70b	0.173	0.405	0.501	0.121

Legend

- Excellent performance (≥ 0.8)
- Good performance (≥ 0.6)
- Average performance (≥ 0.4)
- Below average performance (≥ 0.2)
- Poor performance (< 0.2)
- Missing data

Metrics Explained

- **Correlation:** Pearson correlation between predicted and actual sentiment scores (higher is better)
- **Error:** Mean Absolute Error (MAE) between predicted and actual scores (lower is better)
- **Classification:** Macro-averaged F1 score across sentiment classes (higher is better)
- **Overall:** Combined performance score across all metrics (higher is better)

Summary of Findings

1. **Top performers:** llama_3_3_70b_versatile and claude_3_5_sonnet_single demonstrate exceptional sentiment analysis capabilities, achieving the highest overall scores across all metrics.
2. **Strong contenders:** gpt_4o and gemma2_9b_it also show excellent performance, with particularly strong correlation and classification metrics.
3. **Mid-tier models:** gemini_2_flash_single, deepseek_chat, lsd_en deliver good results, especially considering their smaller size compared to the top performers.
4. **Baseline comparison:** Both LLM models significantly outperform the traditional lexicon-based sentiment detection (LSD) approaches in French and English.
5. **Error analysis:** All models maintain relatively low error rates, with the top performers achieving MAE below 0.2, indicating high accuracy in predicting sentiment intensity.
6. **Classification strength:** Most models excel at the classification task, with macro F1 scores above 0.6 for the top 7 models, showing strong ability to distinguish between negative, neutral, and positive sentiment.